



SCC24 cluster security and administration with Ansible

Matias Zwinger

CSC

Security

In interviews and any written documents, we want to see that you have and use plans, tools, best practices, and documentation to make it through the contest [...]

— SCC24 Systems and Security Evaluation

Documentation

- Write down how you use Ansible on the cluster
- For full points: security plan and incident management
 - Wave your hands a little bit
- Refer to standards
- Use hackmd.io or something similar
 - Make it discoverable for team members

Account management

- No account sharing
- Lock the root account
 - i. e. no one can su root
 - Use sudo
- Don't give out sudo rights without reason
 - The sysadmins can install software dependencies
 - Or better yet, use Ansible to install them
- Have strong password requirements
- Change default passwords
- Disable SSH password login

Configuration management → Ansible

- Do all system administration through Ansible
 - Ansible acts as documentation
- “+4 pts for being able to produce a configuration inventory showing the state of the cluster (and any deviations from planned configuration) on demand”
 - `ansible-playbook with --check and --diff`
- “+2 pts for having different roles based on node function”
 - CPU and GPU nodes in different subgroups

Patching

- Keep your software up to date
- Install security patches if available
 - Just by doing patching you get +3 pts
- Justify any outdated packages in the documentation

Software management

- Install all software using Ansible
- Don't install anything unnecessary
 - Uninstall it if you don't need it
- Install only from trusted sources
 - Package manager
 - Official git repo
 - Official website
 - calculate the checksum!
- +3 pts for being able to list all packages installed

Networking

- Limit attack surface
 - Move SSH to a nonstandard port
 - Be careful to not lock yourself out
 - Install a firewall
 - May be already present on RHEL
- Close management interfaces like IPMI

Logging

- Use a tool like Prometheus to centrally gather logs
- Visualize the data with Grafana
- Set up alerts for possible attacks
 - sshd access log
 - failed sudo attempts

Communication with peers

- “+0.25 pt for each team to which you’ve worked out a way to communicate systems and security tips, tricks, and concerns and actually done so”
→ share your knowledge
- Act on the advice other teams give you
- Report any security issues in the SCC infrastructure to staff

Conclusion

- The judges like standards
 - Justify as much as possible with standards
- Read the whole list of criteria in the Systems and Security Evaluation PDF
 - Sent in the system administration TG chat
 - The appendix has useful links
- Misconfigurations and compromises yield negative points



ANSIBLE

Ansible

- Provisioning tool
- Why not bash scripts?
 - Treat your infrastructure as code!

Example playbook:

1. Make sure user `alice` and is in the groups `user` and `sudo`
2. Install `git`, `vim` and `gcc` using `apt` if they are not installed
3. Make sure a cronjob which runs `foo` every hour exists

- Most actions are idempotent
 - Executing the same playbook 100 times **SHOULD** do the same thing as running it only once.
- Example: The user `alice` exists already
 - Running the playbook again, Ansible will see that `alice` exists and is in the specified groups
 - Exits without modifying anything

```
---                                # Starts the document
# List
- item 1
- item 2
- item 3

dict1:                             # Dictionary
    key1: value1                   # Strings don't need quotation marks
    key2: |
        multi
        line
        string

# List of dictionaries
- subdict1:
    key1: true                     # Boolean
- subdict2:
    key1: no                       # Same as false, but please don't do this
```

```
---
- name: Set up login node users # Names starting with upper case are the standard
  hosts: login-node             # Select which hosts to run on
  become: true                   # Run using superuser rights

tasks:
  - name: Ensure user alice exists
    user:                          # Module name
      name: alice                 # <
      state: present              # < Module arguments
      shell: /bin/bash            # <
      groups: users,sudo          # <

  - name: Install tools
    apt:
      pkg:
        - git
        - vim
        - gcc
```


Running Ansible playbooks

- `ansible-playbook setup.yml`
 - Continue from task foo
`ansible-playbook setup.yml --start-at foo`
 - Dry run
`ansible-playbook setup.yml --check --diff`
 - Run only the tasks marked with tags foo and bar
`ansible-playbook setup.yml --tags foo,bar`
-
- `name: Do something`
 `tags:`
 - `foo`
 - `bar`
 `# [...]`

setup.yml

```
---  
- name: Set up login node users  
  hosts: login-node  
  become: true  
  roles:  
    - setup
```

roles/setup/tasks/main.yml

```
---  
- name: Ensure user alice exists  
  user:  
    name: alice  
    state: present  
    shell: /bin/bash  
    groups: users,sudo  
  
- name: Install tools  
  apt:  
    pkg:  
      - git  
      - vim  
      - gcc
```

```
|─ ansible.cfg
|─ inventory
|  └─ hosts.yml
|─ setup.yml
|─ slurm.yml
|─ hpl.yml
|─ etc.
└─ roles
   └─ slurm
      │   └─ defaults
      │       └─ main.yml
      │   └─ files
      │       └─ install.sh
      └─ tasks
          └─ main.yml
   └─ hpl
      │   └─ tasks
      │       └─ ...
      └─ etc.
```

```
all:
  hosts:
    login-node:
      ansible_host: 1.3.3.7
      ansible_user: sysadmin

  children:
    compute-nodes:
      hosts:
        compute-node-01:
          ansible_host: 10.0.0.1
          ansible_user: root
          ansible_ssh_common_args: '-o ProxyCommand="ssh -W %h:%p -q
sysadmin@1.3.3.7"'
        compute-node-02:
          ansible_host: 10.0.0.2
          ansible_user: root
          ansible_ssh_common_args: '-o ProxyCommand="ssh -W %h:%p -q
sysadmin@1.3.3.7"'
```

Using Ansible for controlling compute nodes

- Ping all compute nodes

```
ansible compute-nodes -m ping
```

- Run a command foo on all compute nodes

```
ansible compute-nodes -m command -a 'foo'
```

- Reboot all compute nodes

```
ansible compute-nodes --become --ask-become-pass -m command -a  
'reboot now'
```

github.com/mk3z/scc24-ansible